

# Stiffness

How to choose  $K$ ?

We need:

①  $K \leq K_{acc}$  (sufficient accuracy)

②  $K \leq K_{stab}$  (absolute stability)

With explicit methods

we need  $K|L| \leq S$

so  $K = \mathcal{O}\left(\frac{1}{|L|}\right)$ .

$K_{acc}$ : depends on how rapidly the solution is changing

$$K = \mathcal{O}\left(\frac{1}{|u'(t)|}\right)$$

## Prothero-Robinson

$$u'(t) = \lambda(u(t) - \cos(t)) - \sin(t)$$

$$u(t_0) = \cos(t_0)$$

$$\lambda < 0$$

Solution:

$$u(t) = \cos(t)$$

For other initial data:  $u(t_0) = \eta$

$$u(t) = \cos(t) + e^{\lambda(t-t_0)}(\eta - \cos(t_0))$$

We say a problem is stiff if:

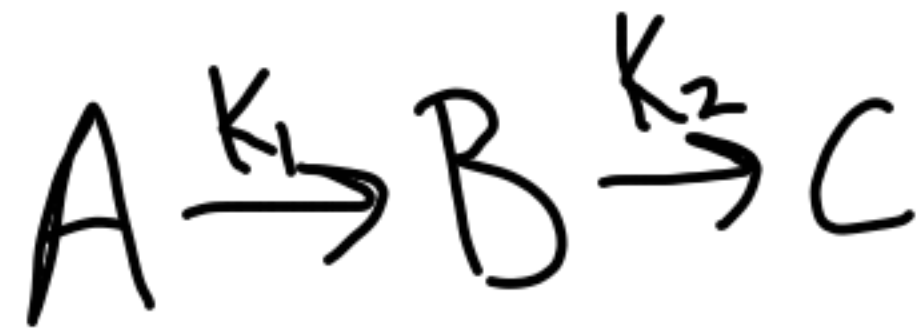
- ① The true solution is slowly changing, but nearby solutions are rapidly changing.
- ② Implicit methods give an accurate solution with a much larger step size than the stable step size for explicit methods. (i.e.  $K_{\text{stab}} \ll K_{\text{acc}}$  for explicit methods)

# Chemical Reaction System

$$u_1'(t) = -k_1 u_1(t)$$

$$u_2'(t) = k_1 u_1(t) - k_2 u_2(t)$$

$$u_3'(t) = k_2 u_2(t)$$



$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}' = \underset{M}{\begin{bmatrix} -k_1 & 0 & 0 \\ k_1 & -k_2 & 0 \\ 0 & k_2 & 0 \end{bmatrix}} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

Eigenvalues of  $M$ :  $0, -k_1, -k_2$

Take  $k_1 \gg 1, k_2 = \mathcal{O}(1)$

Problem is:

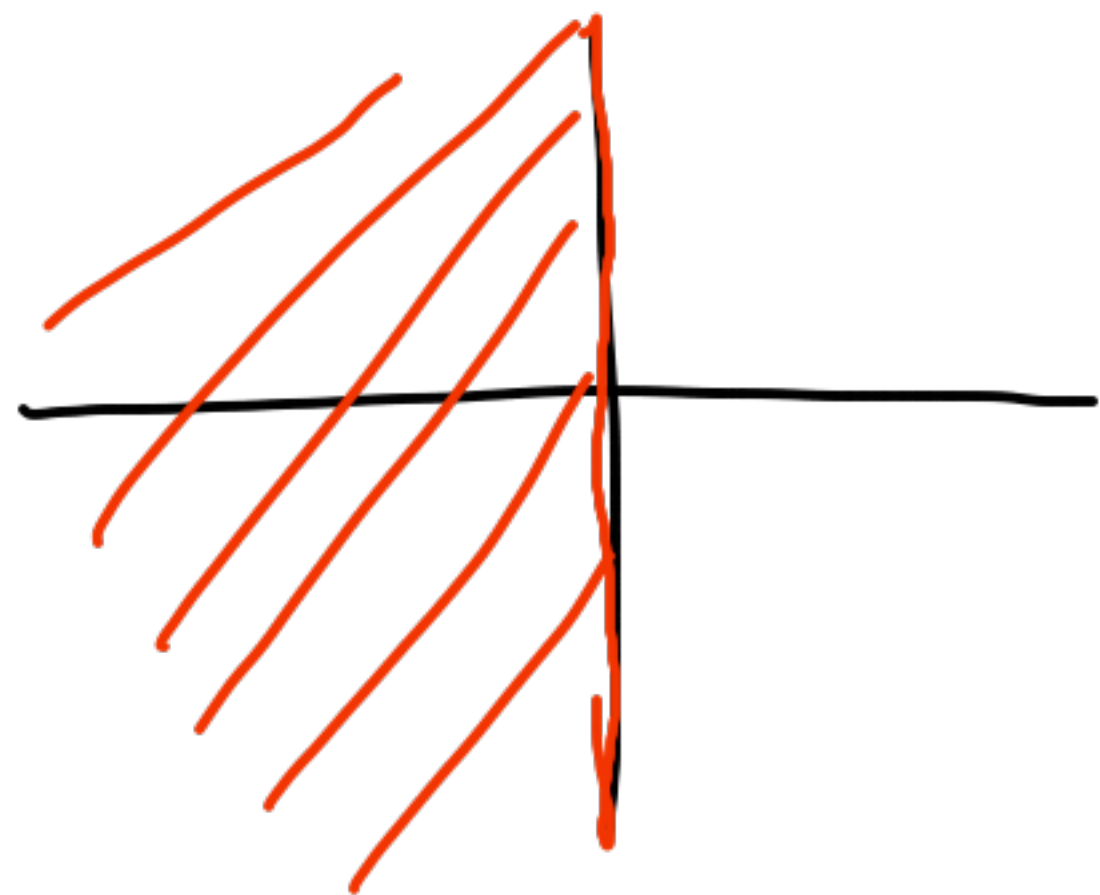
- Not stiff in initial transient phase as  $A \rightarrow B$  (rapidly)
- Stiff later as  $B \rightarrow C$  (slowly)

Absolute stability  
region:

$$S = \{z \in \mathbb{C} : |R(z)| \leq 1\}$$

We say a method is A-stable  
if

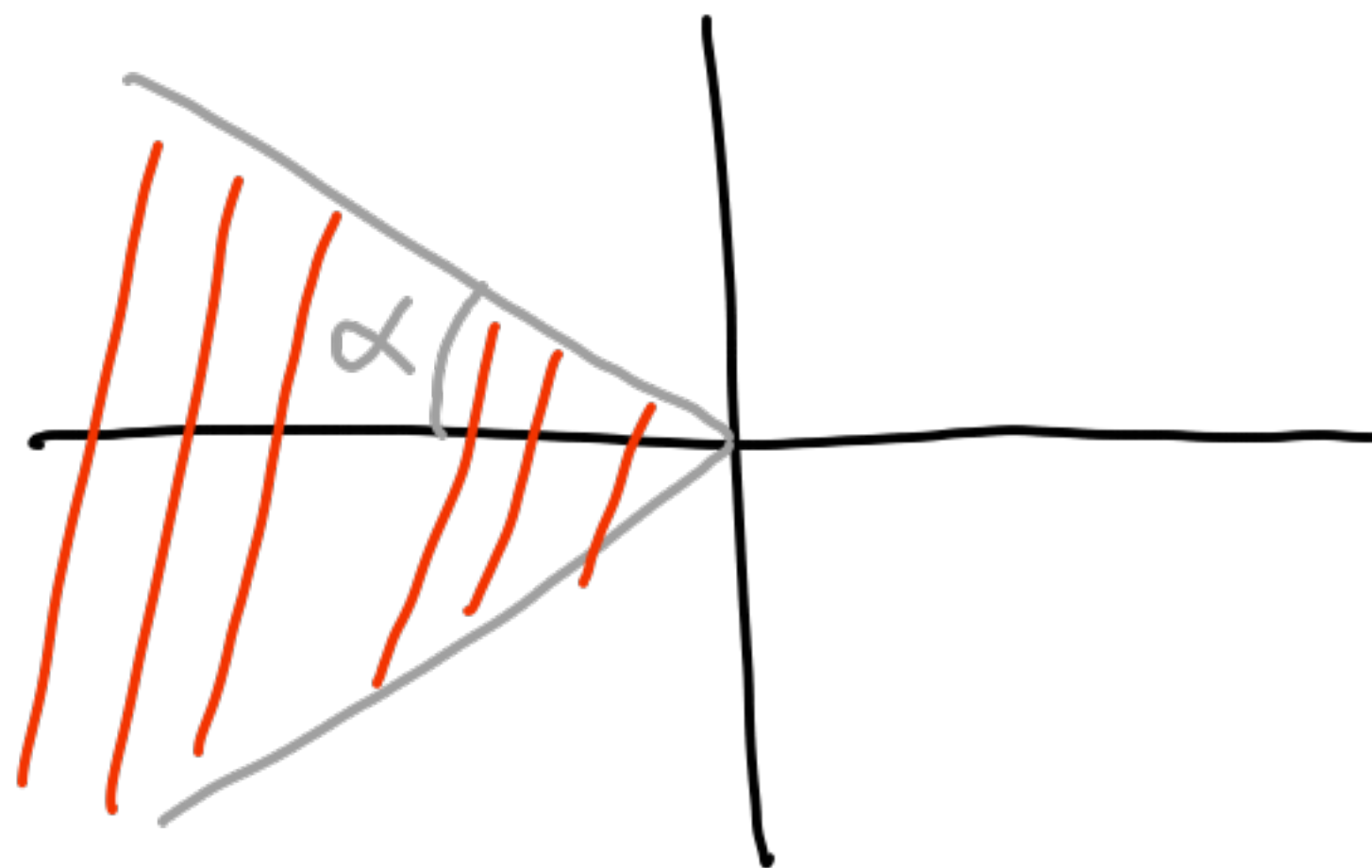
$$\mathbb{C}^- \subset S$$



With an A-stable method,  
 $K_{\text{stab}} = \infty$ , so we can always  
take  $K \approx K_{\text{acc}}$ .

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$A(\alpha)$ -stability



# L-Stability

For  $\lambda \rightarrow -\infty$  the solution of  $u' = \lambda u$  goes to zero arbitrarily fast.

Which implicit methods give the same behavior?

For  $u' = \lambda u$ , with a one-step method,  
we have

$$U^{n+1} = R(z)U^n$$

So we should have

$$\lim_{z \rightarrow -\infty} R(z) = 0 \quad \text{L-stability}$$

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Backward Euler

$$R(z) = \frac{1}{1-z}$$

$$\lim_{z \rightarrow -\infty} R(z) = 0 \quad \checkmark$$

Implicit Trapezoidal  
Method:

$$R(z) = \frac{1 + \frac{z}{2}}{1 - \frac{z}{2}}$$

$$\lim_{z \rightarrow -\infty} R(z) = -1$$

Not L-stable

Good Methods for stiff problems

- BDF (multistep) methods
- Diagonally implicit Runge-Kutta methods